

EDA — schools_combined_enriched_ceeb.csv (English)

Dataset: 25,577 U.S. high schools × 71 columns (24,223 public / 1,354 private, flagged by **sector**), covering 50 states + DC (public includes some territories; private is missing AK and MS).

Data Sources and Time Periods

Variable group	Source	Time period	Granularity	Coverage
School basics, grade counts, race, staffing (public)	NCES CCD (ELSI)	SY 2024-25	School	All public
School basics, religion, coed (private)	NCES PSS	SY 2023-24	School	All private
Graduation rate grad_rate_2021	EDFacts ACGR	SY 2020-21 ⚠ COVID year	School	74% of public
Assessment participation assess_part_*_2021	EDFacts FS185/ FS188	SY 2020-21 ⚠ COVID year; participation, not proficiency	School	77% of public
AP / dual enrollment / SAT-ACT / counselors / certification crdc_*	CRDC	SY 2021-22	School	Public only (AP 51%, counselors 84%)
ISBE report card (ACT/IAR/spending)	ISBE	SY 2024-25 report card (2025 release)	School	IL only (~2.6%)
IB authorization flag	Scraped from IBO	Current as of 2025-26	School	Fuzzy-matched; see caveats
CEEB codes	UC Boulder crosswalk	Compiled (~2024)	School	75% public / 61% private
County income/education/poverty county_* (ACS)	Census ACS 5-year	2020-2024 (released Dec 2025)	County	94%
County child poverty county_pct_child_poverty_saipe	Census SAIPE	2024	County	94%
SAT avg / AP avg / AP per student nu_* (see below)	Bob' s NU Admissions org export	Exported 2026-06; per-metric vintage unstated ⚠ pending Bob	School	~40%

Time alignment: the school roster backbone is 2023-25 while outcome variables fall in 2020-22. Acceptable for cross-sectional clustering, but must be documented; graduation and participation rates were materially affected by COVID and should be interpreted with care.

Key Distributions (by sector)

Variable	Public median	Private median	Notes
Total enrollment	366	223	Right-skewed; public P10=9 (micro/alternative schools), P90=1,735
Pupil/teacher ratio	14.2	10.6	Smaller classes in private
Graduation rate (2021)	90.0	—	Public only; P10=62
AP courses offered	9	—	CRDC, public only
Counselor FTE	2.0	—	P10=0: ~10% of high schools have no dedicated counselor
County median HH income	\$74,467	\$84,488	Private schools sit in wealthier counties
County % bachelor' s+	31.1	36.8	Same pattern
County % child poverty	14.9	13.5	

Private schools are located in systematically wealthier, more educated counties — the community-context dimension will naturally separate sectors in clustering; this is expected.

CEEB Match Quality

	With CEEB	Exact ID match	auto_accept	Needs review
Public	18,263 (75%)	15,079	16,187	2,076
Private	821 (61%)	544	613	208

Missingness Structure

Missingness is structural, not random, in three blocks: ISBE columns are IL-only (97% empty); private-specific columns (religion/coed/PSS ID) exist only for private rows (95% empty); CRDC/EDFacts columns are public-only (AP columns 52% empty — CRDC only collects course counts from schools that offer AP). For modeling, treat the data as public block / private block / shared block rather than dropping rows or mean-imputing globally.

Correlations with Graduation Rate (public)

Assessment participation +0.35, FRL share **-0.34**, enrollment +0.28, SAT/ACT takers +0.26, AP courses +0.23; **county-level income/education are essentially uncorrelated with graduation (± 0.05)** — a school' s internal composition (share of low-income students) matters far more than the average wealth of its county. This supports weighting school-level features over county context in clustering.

Addendum: NU Org Data (schools_org_enriched.csv)

An extended version of the combined table, `csv_exports/schools_org_enriched.csv` (25,577 $\times$ 127), appends 55 `nu_*` columns from the NU Admissions org export (Naviance/ College Board-style). Key metrics: `nu_avg_freshman_sat` (school SAT average, 10,171 schools / 40%, median 1140), `nu_pct_seniors_taking_sat` (9,432, median 58%), `nu_avg_ap_score` (9,841, median 2.7), and `nu_avg_num_ap_tests_taken/_offered` (AP tests per student — directly satisfies Goal 8). Spot check: New Trier shows SAT 1270 / AP 3.87 / 90%+ four-year college — face-valid; SAT correlates 0.61 with AP average.

Usage constraints: coverage is biased toward NU' s recruiting universe (missingness is informative, not random — “no NU data” is itself a signal); text-categorical columns like `nu_percent_going_to_4yr_college` need ordinal encoding; `nu_mean_sat` (11 values) is a dead column; per-metric measurement years are unstated.

Caveats

1. `ib_school_id` is non-null for every private row — the combine step' s fuzzy matcher assigned the nearest candidate to every school. **Always gate on `ib_match_tier`; never treat non-null as “is an IB school.”**
2. Public data includes micro/alternative/virtual schools (P10 enrollment = 9). Apply a minimum-size threshold before clustering (e.g., grades 9-12 enrollment ≥ 30) or bin them separately.
3. Pupil/teacher ratios have outliers >50 (data errors or special settings); winsorize.
4. The `charter` column contains 979 “†” (NCES suppression symbol) that must be converted to NaN.
5. EDFacts values are exam participation rates, not proficiency, and from the COVID year; recommend adding SY 2022-23 Achievement files (FS175/FS178) later.